LAB 2: MEDICAL IMAGE SEGMENTATION

The goal of this lab practice is for the student to become **familiar** with the different basic methods of segmentation of medical images. In particular, we will work with **thresholding, region growing and watershed algorithms,** in addition to introducing **more advanced segmentation algorithms available in the state-of-the-art**.

To check the progress in each of the sections, **you can use the same images used in Lab 1. IMPORTANT: you must indicate the regions of interest (ROI) to be segmented in your images.**

PART 1: THRESHOLDING, REGION GROWING AND WATERSHED

First, **thresholding** is a segmentation technique that separates the pixels of an image according to their gray level. Therefore, it is necessary to correctly select the threshold that allows to separate the sets of gray levels associated with different structures in the image.

On the other hand, the **region growing method** takes as input parameters a set of **seeds** together with the image. These seeds mark the area or objects of the image to be segmented. In this way, the region will grow from these seeds if adjacent pixels meet a certain homogeneity criterion with respect to them.

Finally, the **watershed algorithm** is a morphology-based segmentation technique that allows you to find regions of an image. It is a widely used technique for segmentation and processing of medical images. The basic idea of this algorithm is to consider a numerical image as a topographical relief where the intensity of each pixel corresponds to the height above the relief. The simplest comparation of this segmentation method is to compare it with the flooding of topographical relief (Figure 1), assuming that the water flows of the regional minimums (regional minimum: flat area such that the level of all neighboring pixels is strictly higher).

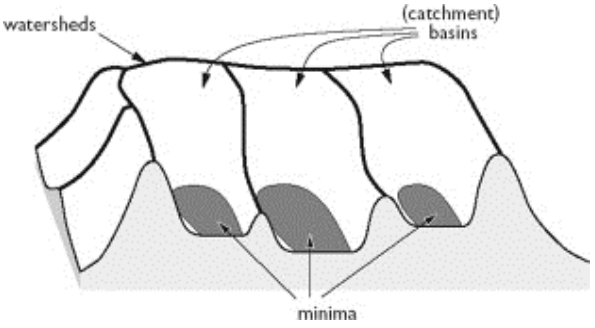


Figure 1 - Comparation with a topographic relief.

Figure 2 shows an example of segmentation by *watershed* in one dimension using the gradient. The segmentation criterion is the homogeneity of intensity levels since watershed determines the zones of influence of the regional minima.

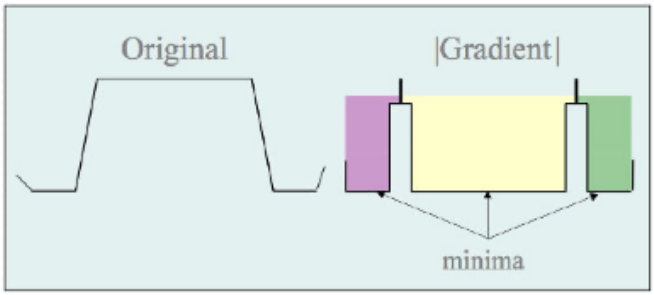


Figure 2 - Example of watershed segmentation in one dimension.

In this exercise you should **evaluate the three presented segmentation algorithms with 2 different medical images containing different anatomical structures.** You should also c**ompare the performance of these segmentation algorithms,** as well as **the influence of their adjustable parameters**. Moreover, you should **justify which segmentation algorithm would be the one of your choice for each anatomical structure (ROI).**

Furthermore, you should d**iscuss whether the previously obtained segmentations could be improved using pre- and post-processing techniques**, such as **histogram modifications, filtering, and/or morphological processing.** You should apply such methodologies on the images you are working with to support your hypothesis.

To do this exercise **you will need to search for implementations of these algorithms on the Internet.**

PART 2: ACTIVE CONTOURS

The segmentation methods exposed are not the only ones that exist, but we can find much more advanced methods: based on active contours, deformable models, or even based on artificial intelligence techniques.

For the latter part of the lab practice it is proposed to resolve the following questions:

- Look for information about one of the most used active contour algorithms: the ***snakes algorithm.*** Briefly explain what it is.
- Evaluate how **it works** in a segmentation scenario. Do it on the previously used images and compare the result of segmenting their ROIs both with snakes and with the algorithms seen throughout this lab practice.

Note: there are several types of *snakes*. In this case, what it is requested is to work on the "simple" active contour method:

- In Python: https://scikit-image.org/docs/dev/auto_examples/edges/plot_active_contours.html.
- In MATLAB: https://www.mathworks.com/help/images/ref/activecontour.html.

PART 3: OTHER SEGMENTATION METHODS

In this last exercise you will have to conduct a search for **other types of segmentation methods** that are available in the state-of-the-art. You will have to perform an in-depth explanation of a method you want, search for the code implementation (or implement it if the code is not available), evaluate the results of the selected algorithm with the same images as in the previous exercises, and **compare its performance with previously used segmentation algorithms.**

SUBMISSION AND GRADING

For this lab practice it is requested to submit a ***.zip* file** with the following items:

- A **single PDF** containing the **report** of the completed assignment, with a **maximum length of 10 pages**. The methods used must be justified and the results obtained must be discussed, as indicated in the exercise statements.
- The **source code developed and used** to complete each exercise. You can use either Python or MATLAB.
- **The images used** throughout the assignments.

Since the assignment will be completed in pairs, the **report must clearly indicate the contribution of each team member**, **without prejudice to any additional checks that the professors may carry out** to ensure that all members have acquired the competencies expected from this practical assignment. In addition, the **use of AI tools** must be described if they have been used (i.e., in which sections and for what purposes). AI can be a useful tool for **assisting with code implementation**, but it **MUST NOT replace the students' programming work or judgment, NOR SHOULD IT BE USED for the interpretation or discussion of the results**.

Please remember that this is one of the last courses you need to pass before becoming a Biomedical Engineering graduate. Therefore, you should approach this assignment as if you were working in a hospital or a company and had to present it as part of your professional work. Consequently, the **final grade** will be based on the quality of both, the **implemented or utilized code** (including appropriate explanatory comments) and the **report (including the explanation of the methods, their parameters and how they affect image segmentation, as well as the analysis and discussion of the obtained results)**.

The **.zip file to be submitted** must be named as follows: **P2_MIA_BED_26_27_GroupX.zip,** where *GroupX* is the name of the group (for example, **P2_MIA_BED_26_27_G1_2.zip** would correspond to the submission of group **G1_2**).

The **submission deadline** is **November 5, 2026, at 23:55** for students carrying out the laboratory session on **October 22, 2026** (Groups **G1_x**), and **November 6, 2026, at 23:55** for students carrying out the laboratory session on **October 23, 2026** (Groups **G2_x**).